

TASK 2: API INTEGRATION AND JSON HANDLING

Objective:

Learn how Python communicates with external APIs and handles JSON data.

API script:

```
import requests

from datetime import datetime

print("==== WEATHER INFORMATION USING API =====")

# Take city name input

city = input("Enter city name: ")

# OpenWeatherMap API URL

url=f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid=#yourapikey&units=metric"

try:

    # Sending GET request

    response = requests.get(url, timeout=10)

    # Convert response into JSON

    data = response.json()

    # Check successful response

    if response.status_code == 200:

        # Extract required information

        city_name = data["name"]

        country = data["sys"]["country"]

        temperature = data["main"]["temp"]

        feels_like = data["main"]["feels_like"]

        humidity = data["main"]["humidity"]

        pressure = data["main"]["pressure"]

        weather = data["weather"][0]["description"]

        wind_speed = data["wind"]["speed"]

        visibility = data.get("visibility", 0) / 1000 # meters to km
```

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sunrise_timestamp = data["sys"]["sunrise"]
sunset_timestamp = data["sys"]["sunset"]

# Convert timestamps into readable time
sunrise = datetime.fromtimestamp(sunrise_timestamp).strftime("%I:%M:%S %p")
sunset = datetime.fromtimestamp(sunset_timestamp).strftime("%I:%M:%S %p")

# Current Date and Time
current_time = datetime.now().strftime("%d-%m-%Y %I:%M:%S %p")

# Filtering logic
if temperature > 30:
    climate = "Hot"
elif temperature > 20:
    climate = "Warm"
else:
    climate = "Cool"

# Display output
print("\n===== WEATHER REPORT =====")
print("Date & Time:", current_time)
print("\nLocation Details")
print("City:", city_name)
print("Country:", country)
print("\nWeather Details")
print("Temperature:", temperature, "°C")
print("Feels Like:", feels_like, "°C")
print("Humidity:", humidity, "%")
print("Pressure:", pressure, "hPa")
print("Weather Condition:", weather)
print("Climate Type:", climate)
print("\nWind & Visibility")
print("Wind Speed:", wind_speed, "m/s")
print("Visibility:", visibility, "km")

```

```

    print("\nSun Timings")

    print("Sunrise:", sunrise)

    print("Sunset:", sunset)

else:

    print("\nInvalid city name or API key.")

    print("Error Message:", data["message"])

# Connection Error

except requests.exceptions.ConnectionError:

    print("Internet connection error.")

# Timeout Error

except requests.exceptions.Timeout:

    print("Request timed out.")

# General Request Errors

except requests.exceptions.RequestException as e:

    print("Error:", e)

print("\n===== PROGRAM COMPLETED =====")

```

Explanations/Concepts Learned:

- **API Integration**
Learned how to connect Python applications with external services using APIs to fetch real-time data.
- **JSON Parsing**
Understood how JSON data is structured and how to extract required information such as temperature, humidity, and weather conditions using Python dictionaries and lists.
- **HTTP Requests**
Learned how HTTP GET requests work and how Python communicates with web servers to request data.
- **Python Requests Library**
Used the requests library to send API requests and receive responses efficiently in Python.
- **Error Handling**
Implemented error handling techniques to manage invalid city names, failed requests, and unexpected API responses without crashing the program.
- **Real-time Data Fetching**
Learned how to retrieve and display live weather information dynamically from the OpenWeatherMap API.

Outputs:

```
E:\alfido py dev\task2>python index.py
===== WEATHER INFORMATION USING API =====
Enter city name: Hyderabad

===== WEATHER REPORT =====
Date & Time: 22-05-2026 04:34:32 PM

Location Details
City: Hyderabad
Country: IN

Weather Details
Temperature: 38.23 °C
Feels Like: 40.57 °C
Humidity: 32 %
Pressure: 1003 hPa
Weather Condition: haze
Climate Type: Hot

Wind & Visibility
Wind Speed: 7.2 m/s
Visibility: 5.0 km

Sun Timings
Sunrise: 05:42:31 AM
Sunset: 06:43:15 PM

===== PROGRAM COMPLETED =====
```

```
Command Prompt

E:\alfido py dev\task2>python index.py
===== WEATHER INFORMATION USING API =====
Enter city name: delhi

===== WEATHER REPORT =====
Date & Time: 22-05-2026 04:35:05 PM

Location Details
City: Delhi
Country: IN

Weather Details
Temperature: 43.05 °C
Feels Like: 41.82 °C
Humidity: 15 %
Pressure: 999 hPa
Weather Condition: haze
Climate Type: Hot

Wind & Visibility
Wind Speed: 4.63 m/s
Visibility: 5.0 km

Sun Timings
Sunrise: 05:27:03 AM
Sunset: 07:08:48 PM

===== PROGRAM COMPLETED =====
```

Github repository: https://github.com/vineelkaya-create/AlfidoTech_task2/blob/main/index.py

